EX.No.2.1 – CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

DATE:12.10.2026

AIM:

To create a 3D solid model from the given isometric drawing using Fusion 360 software.

REQUIREMENTS:

1. Autodesk Fusion 360
2. Windows 10 operating system
3. 16 GB RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download and 500 Kbps or faster upload speed

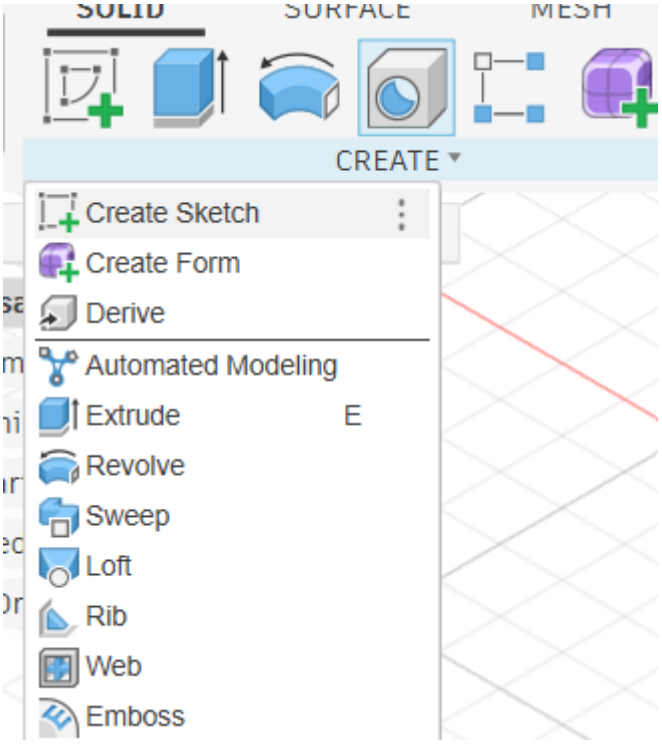
PROCEDURE:

STEP 1: Open Fusion 360

Launch Autodesk Fusion 360 and create a new design file.

STEP 2: Create a Sketch

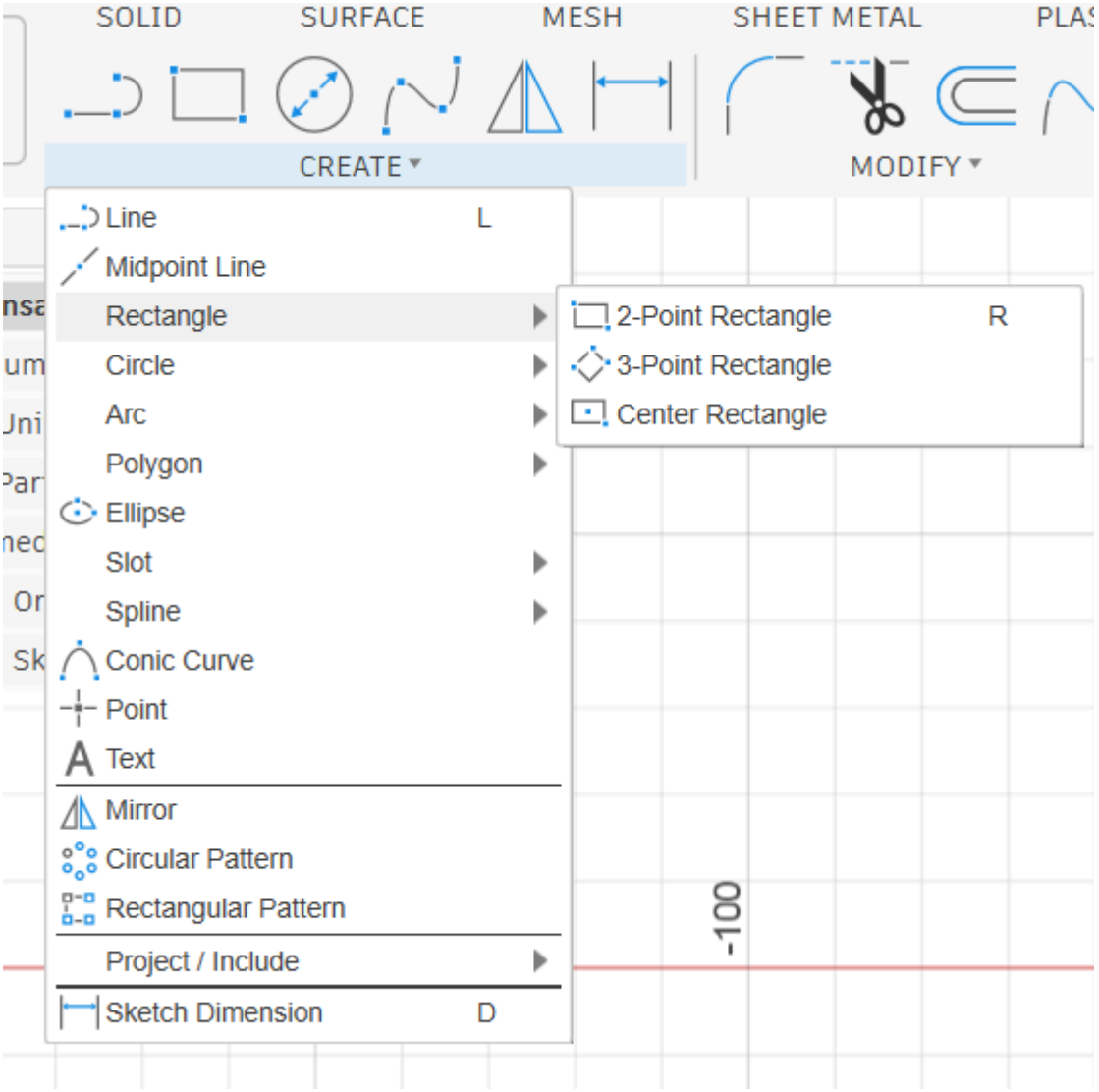
Click on **Create Sketch** and select the Front (XZ) plane as the sketching surface.



STEP 3: Draw the Base Rectangle

Choose **Rectangle → Center Rectangle** from the Sketch menu. Place the rectangle at the origin and specify its width as 100 mm and height as 60 mm. Click Finish Sketch to complete

the base profile.



STEP 4: Create the Base Solid

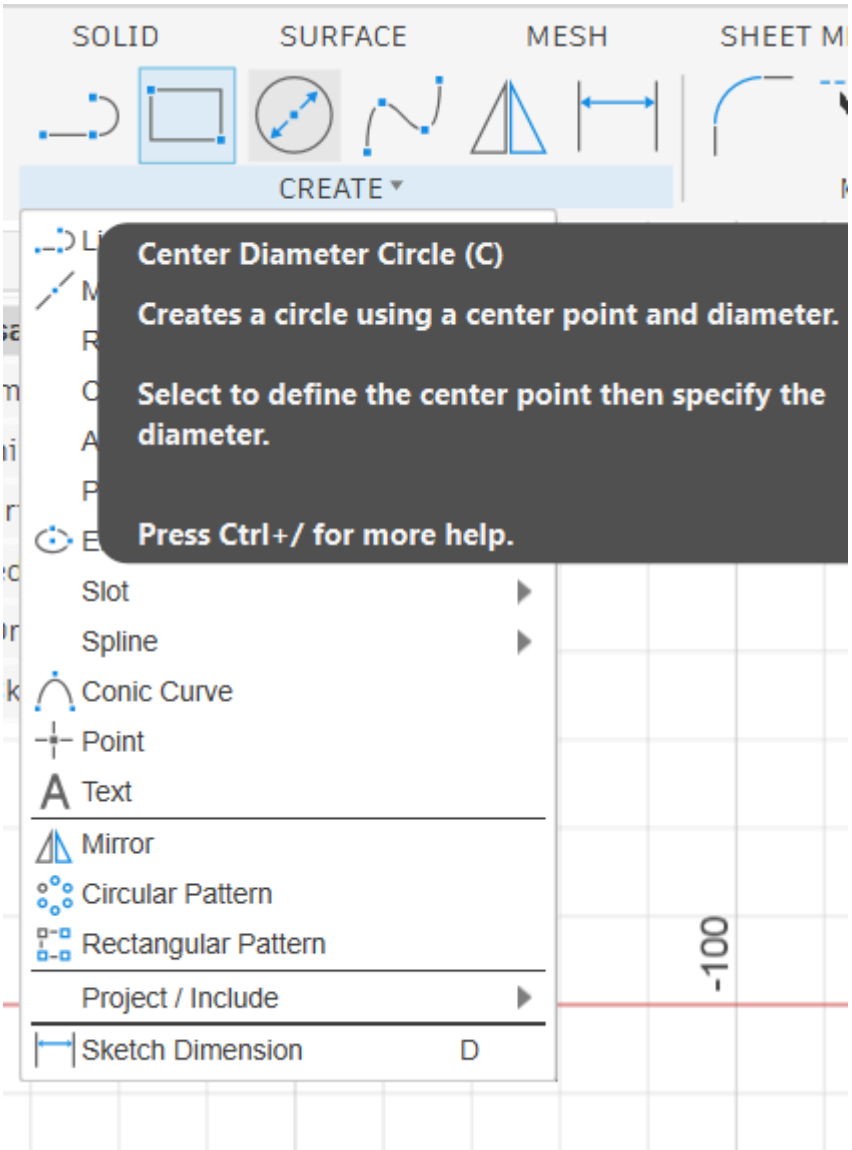
Select the rectangle profile and use the **Extrude** command. Enter 30 mm as the extrusion distance and press Enter to form the rectangular solid.

STEP 5: Sketch on the Front Face

Select the front face of the rectangular solid and click Create Sketch to draw the next feature.

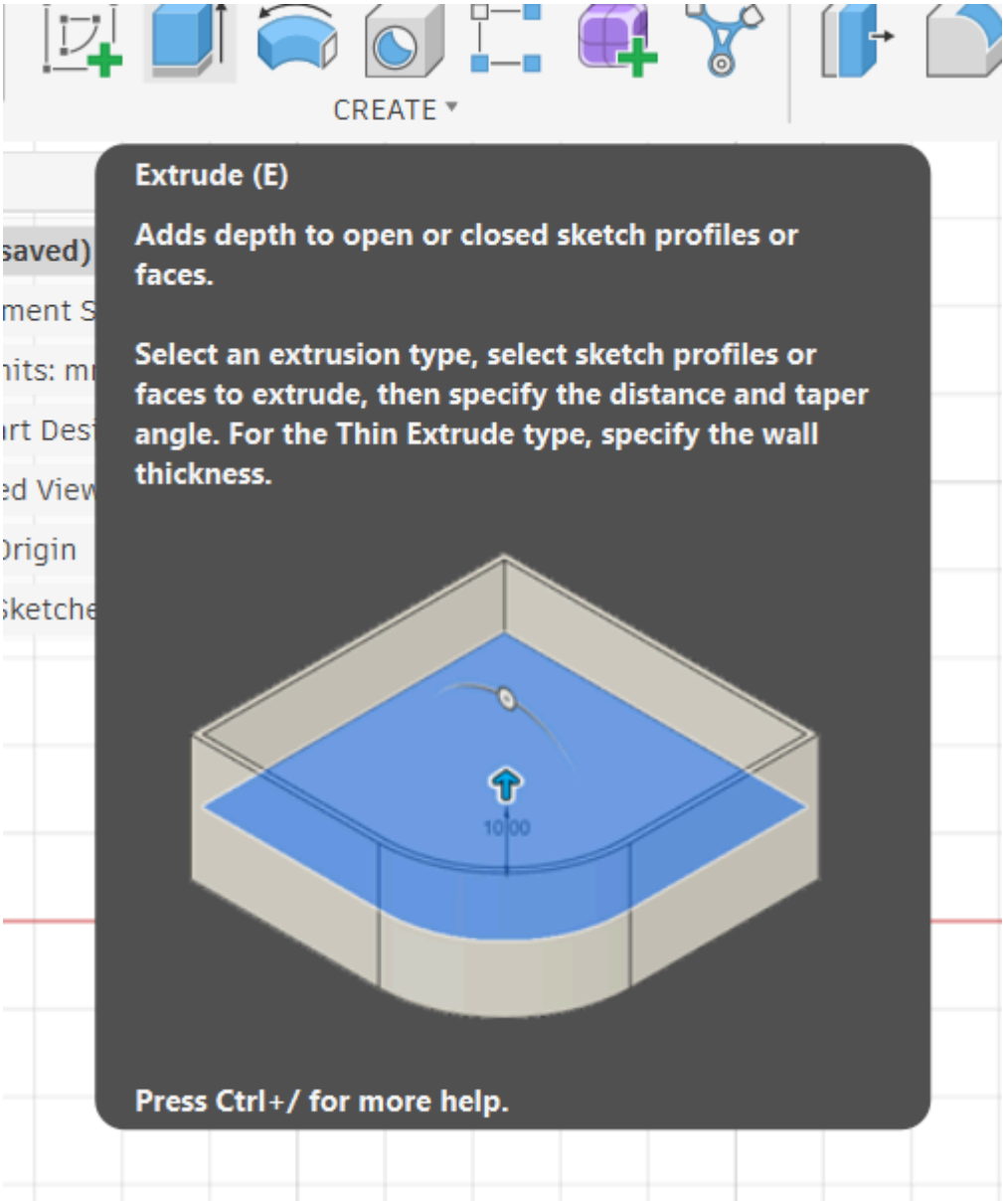
STEP 6: Draw the Circle

Choose **Circle** → **Center Diameter Circle** and place the circle at the required centre. Set the diameter to 100 mm and click Finish Sketch.



STEP 7: Extrude the Circular Profile

Select the circular profile and activate the Extrude command. Enter 20 mm as the extrusion distance and press Enter to create the circular feature.



STEP 8: Create a Sketch on the Top Face

Select the top surface of the rectangular block and start a new sketch.

STEP 9: Draw the Required Lines

Use the Line command to draw the required lines and projections according to the given isometric drawing. Ensure that the sketch matches the required shape and dimensions.

STEP 10: Complete the Final Feature

Click Finish Sketch, select the required closed profile and use the Extrude command wherever necessary to form the remaining solid features.

STEP 11: Verify the Model

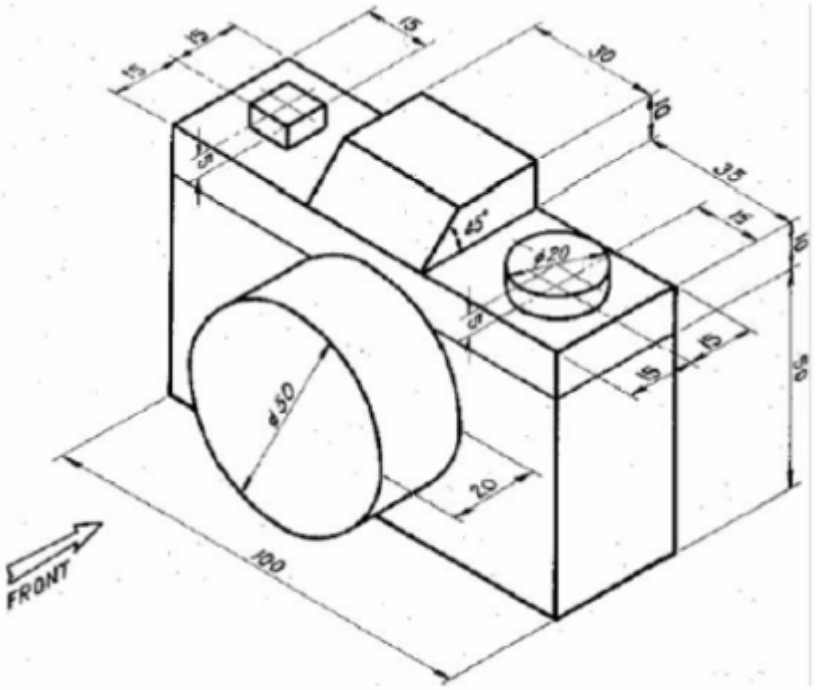
Check the completed 3D model from different viewing angles. Compare it with the given isometric drawing and make any necessary corrections.

STEP 12: Save the Design

Save the completed model in Fusion 360 with a suitable file name.

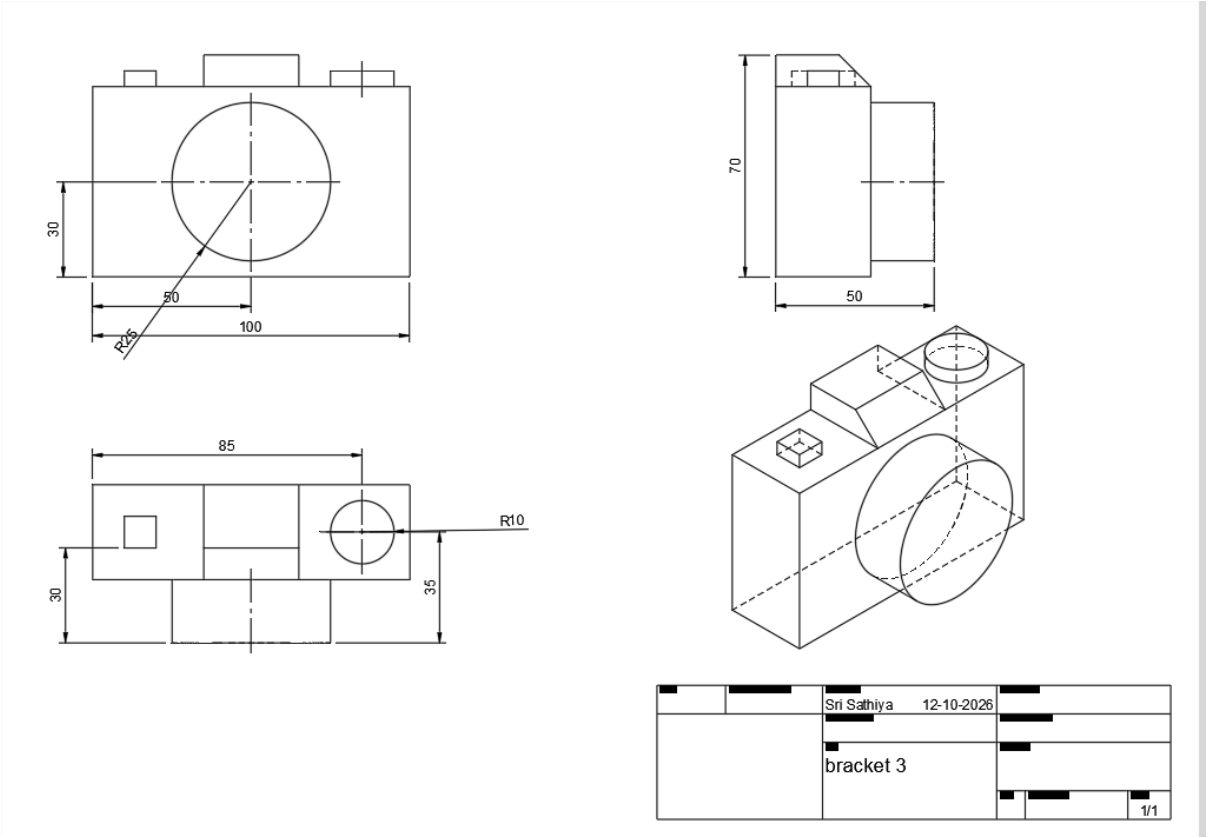
INPUT:

Given isometric drawing.



OUTPUT:

3D solid model created using Autodesk Fusion 360.



RESULT:

Thus, the 3D solid model was created from the given isometric drawing using Autodesk Fusion 360 software.